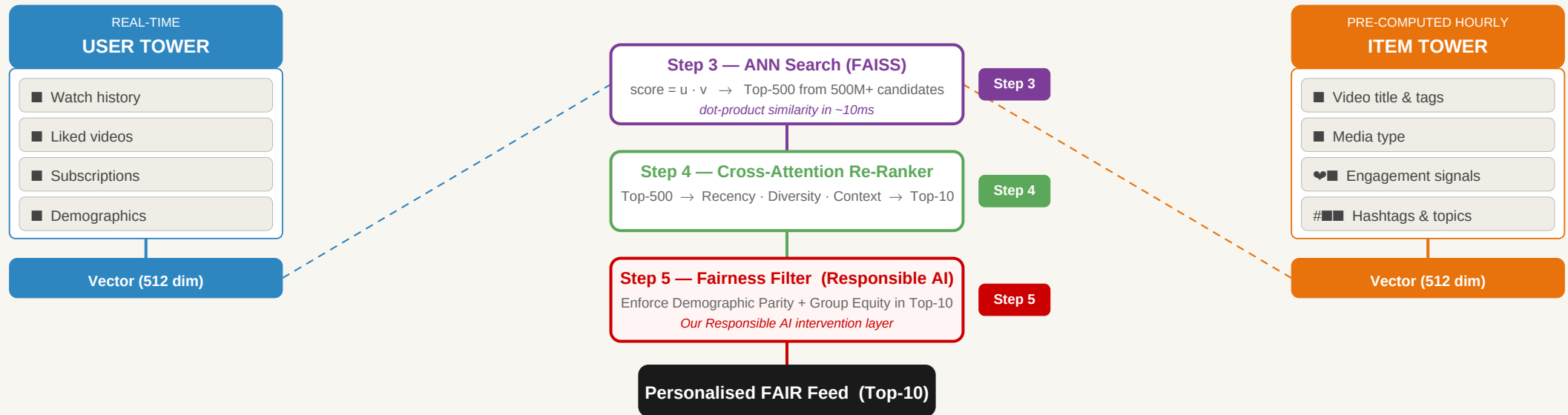


Two-Tower Transformer — How YouTube Recommends Your Next Video



LEGEND: ■ User Tower ■ Item Tower ■ ANN Search ■ Re-Ranker ■ Fairness Filter

Step 1 — User Tower

YouTube converts YOUR watch history, likes & subscriptions into 512 numbers = your fingerprint.

Step 2 — Item Tower

Every video gets converted into 512 numbers based on its title, tags & engagement. Done in advance.

Step 3 — ANN Matching

Your 512 numbers are matched against all video fingerprints. Best 500 found in milliseconds.

Step 4 — Re-Ranker

Top 500 re-scored using deeper signals: recency, diversity, session. Narrowed to Top 10.

Step 5 — Fairness Filter

Checks if any group is under-shown. Adjusts ranking to ensure fair representation. Our RAI fix.

Why Two Towers?

800M videos x 50M users = 40 quadrillion comparisons/day. Pre-computing item vectors makes matching take only ~10ms.